Applied Soft Computing 110 (2021) 107607

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Deep reinforcement learning and parameter transfer based approach for the multi-objective agile earth observation satellite scheduling problem

Luona Wei, Yuning Chen *, Ming Chen, Yingwu Chen

College of Systems Engineering, National University of Defense Technology, Changsha, 410073, China

ARTICLE INFO

Article history:
Received 26 November 2020
Received in revised form 21 May 2021
Accepted 8 June 2021
Available online 17 June 2021

Keywords:
Agile satellite scheduling
Multi-objective
Deep reinforcement learning
MOEA

ABSTRACT

The agile earth observation satellite scheduling problem (AEOSSP) consists of selecting and scheduling a number of tasks from a set of user requests in order to optimize one or multiple criteria. In this paper, we consider a multi-objective version of AEOSSP (called MO-AEOSSP) where the failure rate and the timeliness of scheduled requests are optimized simultaneously. Due to its NP-hardness, traditional iterative problem-tailored heuristic methods are sensitive to problem instances and require massive computational overhead. We thus propose a deep reinforcement learning and parameter transfer based approach (RLPT) to tackle the MO-AEOSSP in a non-iterative manner. RLPT first decomposes the MO-AEOSSP into a number of scalarized sub-problems by a weight sum approach where each sub-problem can be formulated as a Markov Decision Process (MDP). RLPT then applies an encoder–decoder structure neural network (NN) trained by a deep reinforcement learning procedure to producing a high-quality schedule for each sub-problem. The resulting schedules of all scalarized sub-problems form an approximate pareto front for the MO-AEOSSP. Once a NN of a subproblem is trained, RLPT applies a parameter transfer strategy to reducing the training expenses for its neighboring sub-problems. Experimental results on a large set of randomly generated instances show that RLPT outperforms three classical multi-objective evolutionary algorithms (MOEAs) in terms of solution quality, solution distribution and computational efficiency. Results on various-size instances also show that RLPT is highly general and scalable. To the best of our knowledge, this study is the first attempt that applies deep reinforcement learning to a satellite scheduling problem considering multiple objectives.

1. Introduction

The earth observation satellites (EOS) plays a significant role as a space information platform, observing ground targets, collecting images and data in accordance to requests from all kinds of users. The EOS has been widely used and has become increasingly effective in providing information support for applications among disaster monitoring, emergency response, environmental protection, traffic prediction and so on. Compared with the conventional EOS (CEOS), the agile earth observation satellite (AEOS), a new generation of EOS, appears to have superior attitude maneuver ability, which can rotate rapidly and flexibly in roll, pitch and yaw axes. Thus, the AEOS is able to observe targets in prolonged visible time windows (VTWs). Furthermore, more complicated imaging requirements such as stereo imaging and a set of geographically close tasks can be accomplished by the AEOS owing to its advanced maneuverability.

The AEOS scheduling problem (AEOSSP, shown in Fig. 1) is very hard to solve due to its enhanced capability, especially when compared with the CEOS scheduling problem (CEOSSP) which is naturally more regarded as a selecting problem. The AEOSSP is a highly time-dependent problem, i.e., a slight modification of the start-time for an observation may lead to bidirectional propagated changes towards the transition time required for maneuvering. Consequently, the AEOSSP is of greater complexity in comparison with the CEOSSP because of its larger search space.

The AEOSSP is proved to be a NP-hard problem [1–3]. Intensive research has been carried out during the past years. Lemaitre et al. [4] defined the general description of the AEOSSP, analyzing its characteristics and complexity, then strategies including a greedy constructive algorithm, a dynamic programming algorithm, a constraint programming approach and a local search method were applied to solving the proposed problem. Bianchessi et al. [5] solved the scheduling problem of multi-satellites performing multiple orbits by describing a tabu search heuristic and the quality of the solutions was evaluated by an upper bounding procedure. Habet et al. [6] formulated the AEOSSP

* Corresponding author.
E-mail addresses: wlnelysion@163.com (L. Wei), cynnudt@hotmail.com (Y. Chen), chenming_nudt@163.com (M. Chen), ywchen@nudt.edu.cn (Y. Chen).

https://doi.org/10.1016/j.asoc.2021.107607
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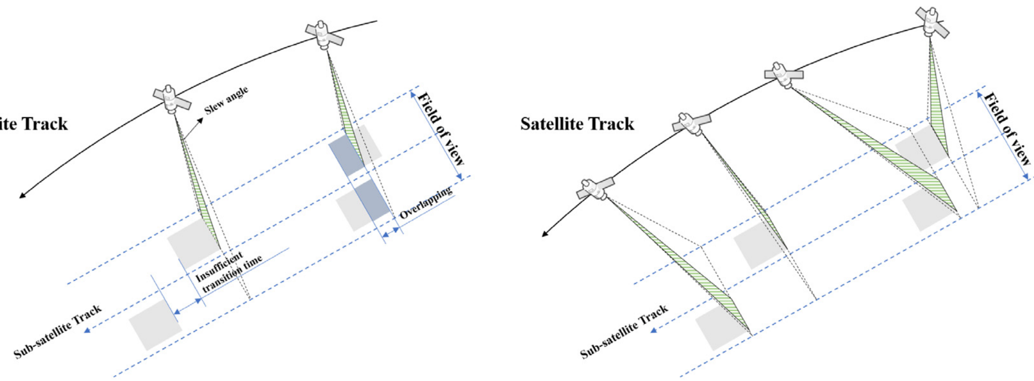


(a) The observations executed by the CEOS (b) Scheduling conflict targets by solving the AEOSSP

Fig. 1. Illustration of the AEOSSP. The AEOSSP aims to select and schedule given requests to satisfy the optimization goal. By addressing the AEOSSP, consecutive requests which cannot be observed by the same CEOS because of the overlapping or the insufficient transition time (as (a) shows) can be imaged by an AEOS owing to the capability of pitch, yaw and roll axes (shown in (b)).

as a constrained optimization problem which involves stereoscopic and time window visibility constraints, then proposed a tabu search algorithm hybridized with a systematic search to find solutions among a wide-range and efficient search space. Wang et al. [7] presented a nonlinear model with a priority-based heuristic considering conflict-avoided, limited backtracking and downloading, and a decision support system was also proposed. Besides above mentioned research, a variety of other metaheuristic or heuristics have been applied, including simulated annealing [8], tabu search [9], genetic algorithms [10,11], evolutionary algorithms [12]. Liu et al. [13] developed an adaptive large neighborhood search (ALNS) which consisted of six removal operators and three insertion operators for producing solutions, considering time-dependent transition time of an AEOSSP. He et al. [14] extended the ALNS, applying it at the multi-satellite case, and proposed an adaptive-ALNS(A-ALNS) which included an adaptive task assignment mechanism to guide searching.

Besides the original AEOSSP which usually takes scheduled observation profits as objective function, AEOSSP with multiple objectives has also attracted the interest of researchers. Tangpattanakul et al. [15] introduced an indicator-based multi-objective local search for the AEOSSP and took maximizing the total profit and minimizing the maximum profit difference between users as optimization targets. Wang et al. [16] established a multi-objective integer programming model by considering both the mission planning and the data transmission process, then the Strength Pareto Evolutionary Algorithm II(SPEA2) was utilized to maximize the total profit and minimize the energy consumption simultaneously. Sun et al. [17] gave four criteria as objectives to evaluate scheduling results and designed a SPEA2 with individual coding and decoding strategies. Li et al. [12,18] considered the AEOSSP with three and more than three objectives respectively, then adopted several preference-based evolutionary multi-objective optimization (EMO) methods in various scheduling scenarios.

Previous studies basically apply EMO and local search methods to deal with the multi-objective AEOSSP (MO-AEOSSP). However, there are several inevitable weaknesses for these iteration-based algorithms. First, most operators and decision methods have to be considered by the designer and the coding and decoding processes may become rather complicated for problems with many constraints [19]. The parameter settings, e.g. the probability of crossover and mutation, also highly rely on expert experience. Impropriate parameters may lead to unpredictable effect towards the quality of solutions [20]. Then, it may be quite time-consuming for approximating the Pareto-Front (PF) of a MO-AEOSSP, especially large-scale one, since a large number of iterations and evaluations are required and constraint checking is needed in every generation [21]. In addition, once there is any change of the problem, the algorithm must be re-executed to get new solutions. In other words, an algorithm is always designed for a scientific scenario or instance that even a similar scenario may cause a drastic revision of the solver to obtain a good result [22].

To solve the above issues, researchers begin to seek for new non-iterative methods without hand-engineered reasoning. With the advanced development of neural network and deep reinforcement learning (DRL), it occurs to researchers that NP-hard combinatorial optimization can be formulated as sequential decision-making tasks due to their highly-structured nature [23], and DRL based methods can be adopted to replace handcrafting solutions. Oriol Vinyals et al. [24] first proposed a Pointer Network, using attention mechanism to predict the city permutation of traveling salesman problem (TSP). MohanmmadReza Nazeri et al. [25] presented an end-to-end framework, using reinforcement learning to solve a Vehicle Routing Problem. Li et al. [26] solved multi-objective TSPs (MOTSPs) by proposing an end-to-end framework named DRL-MOA, Pointer Network and a uni-directional neighborhood-based parameter transfer strategy are adopted in their work. Peng et al. [27] developed a sequential decision-making model to schedule the satellite observation task in real-time, a long short-term memory-based encoding network and a classification network are incorporated into the deep learning based planning method. In summary, all the above studies have indicated that DRL is capable of solving (multi-objective) combinatorial optimization problems like the MO-AEOSSP and can overcome most of the aforementioned disadvantages which exist in iteration-based methods.

To provide a more intuitive view, Table 1 is presented to classify the reviewed literature with different solving methods toward the AEOSSP and MO-AEOSSP. From the table we can figure out that most of the researchers addressed the MO-AEOSSP by metaheuristic methods. Some machine learning methods have been adopted on the research of the single-objective AEOSSP recently. However, the MO-AEOSSP has never been addressed through a DRL method or a machine learning method until now.

Inspired by recent works towards DRL-based combinatorial optimization, we propose a deep reinforcement learning and parameter transfer based approach (RLPT) to handle the MO-AEOSSP in a non-iterative manner. In this paper, the failure rate and the timeliness of all scheduled requests are taken as the two objectives to be optimized. The MO-AEOSSP is first divided into a series of sub-problems by a weight sum approach, then each sub-MO-AEOSSP is viewed as a Markov Decision Process and solved by

Table 1
Overview of the applied methods in previous research on various AEOSSP.

Method	Formulation characteristic	
	Single-objective AEOSSP	MO-AEOSSP
Exact method	[28], [29], [30], [31], [32], [33], [34]	–
Heuristic	[35], [4], [36], [37], [7], [38], [39], [40], [41]	–
Metaheuristic	[8], [10], [42], [43], [44], [4], [13], [14]	[15], [11], [12], [18], [45], [46], [47], [48], [17]
Machine learning	cooperative neuro-evolution [3], LSTM [27] reinforcement learning [22]	–

an encoder–decoder structured neural network which is trained by an Actor–Critic method. Parameter transfer strategy is adopted to enhance the efficiency for training in order to shorten the total training cost. Computational experiments show that the resulting neural networks provided by the DRL procedure can effectively address the studied MO-AEOSSP. RLPT is able to produce better quality approximate pareto front in comparison with several classical multi-objective evolutionary algorithms (MOEAs) and also outperforms them in terms of the hypervolume and computing time indicator.

The proposed RLPT is appealing since it can provide high-quality solutions for the MO-AEOSSP by sequential decision-making and feed-forward procedures rather than a large number of iterations along with coding and decoding processes, which is always regarded as the main shortcomings of many heuristic and metaheuristic algorithms. Besides, RLPT is of great scalability. For classical algorithms, changes of instance sizes or even a slight modification of a target attribute may lead to long-time re-execution. But RLPT can automatically adapt the solution without re-training owing to its Markov property and self-driven learning mechanism.

In conclusion, the main contributions of this work are as follows:

1. The development of the RLPT for the MO-AEOSSP, offering a new non-iterative way for solving this practical problem.
2. The mathematical model and two reasonable optimization objectives of MO-AEOSSP are described. A dataset towards the MO-AEOSSP is created, which is publicly available for related research.
3. Experimental results show that the proposed RLPT outperforms standard MOEAs with competitive performance in terms of convergence, spreading and computing efficiency.

The reminder of the paper is organized as follows. In Section 2, the MO-AEOSSP is described and the mathematical model is presented. In Section 3, RLPT is introduced in detailed and its application for solving the MO-AEOSSP is presented. Computational studies are presented in Section 4. Lastly, some discussion and conclusions are drawn in Section 5.

2. Multi-objective agile earth observation satellites scheduling problem (MO-AEOSSP)

Based on previous studies, the AESSOP can be described as an oversubscribed problem which aims to select a subset of candidate tasks from user requests and to decide when to observe those chosen tasks under the condition of meeting all the constraints. In this paper, two reasonable objectives, request failure and timeliness, are chosen as the optimization targets of a MO-AEOSSP. In this section, several concepts and assumptions are introduced and the mathematical model of the MO-AEOSSP is established.

2.1. Problem description and assumptions

In this work, a multi-objective version of AEOSSP is considered, which takes more than one objectives as optimization goals in order to meet different demands rather than simply maximizing the total profits of selected requests as many previous works did.

Since this paper mainly focuses on the multi-objective optimization of MO-AEOSSP, several reasonable assumptions and simplifications are made as follows:

- The observation satellite has sufficient onboard power.
- The image download process is not taken into account.
- All the tasks are point targets or small areas that can be observed in one pass. Polygon and stereo targets are not considered in this study.
- At any time only one imaging request can be shot and once an observation task is executed, it cannot be preempted by another request.
- Only the single-orbit scenario of single satellite is considered.

2.2. Notations and variables

Given a set of candidate requests as inputs, denoted by $R = \{r_1, \ldots, r_N\}$, where N is the number of requests to be scheduled, each request $r_i \in R$, is defined as follows:

i: the identifier for r_i;

p_i: the priority of r_i, measuring the profit earned if r_i is observed;

du_i: the required duration for observing r_i;

$tw_i = [tw_i^s, tw_i^e]$: the VTW of r_i, tw_i^s represents the start-moment of tw_i while tw_i^e is the end moment of tw_i;

tr_{ij}: the transition time for maneuvering from last observation mission r_i to execute r_j;

M: the maximum total time of observations;

$[ST, ET]$: the scheduling horizon of the MO-AEOSSP.

The scheduling of AEOS needs not only to acquire a subset of weighted observation tasks, but also to determine the exact start-moment for every selected request within their VTW. Therefore, here we define the following decision variables in this study:

$$x_i = \begin{cases} 1, \textit{the request } r_i \textit{ is selected} \\ 0, \textit{otherwise} \end{cases}$$

t_i: a continuous variable which denotes the start-time of observing r_i if r_i is selected.

2.3. Objectives

Two objectives are proposed in this section to formulate a bi-objective function of the MO-AEOSSP:

$$\min F = \{f_1, f_2\}$$

$$f_1 = 1 - \sum_{i=1}^{N} x_i \cdot p_i \Big/ \sum_{i=1}^{N} p_i$$

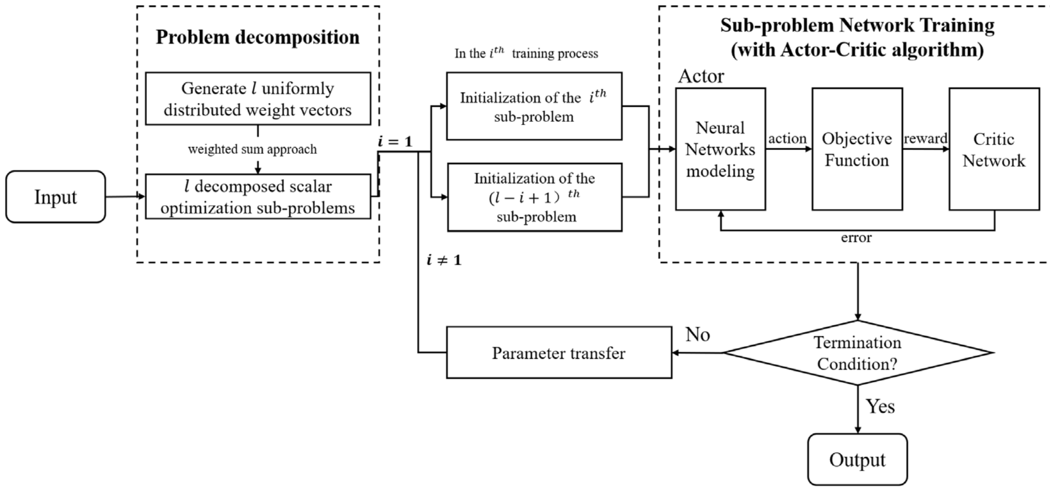


Fig. 2. The framework of RLPT. The input problem is decomposed into l sub-problems through l uniformly generated weight vectors. i denotes the index of the training process. At the first training process ($i = 1$), initial trainable parameters of the networks for the first and the last sub-problems are randomly generated. For the others ($i \neq 1$), the initial trainable parameters for the networks are set though a parameter transfer strategy. Neural network for each sub-problem is modeled and trained through the **Sub-problem Network training** procedure.

$$f_2 = \frac{1}{N}\sum_{i=1}^{N} x_i \cdot \frac{t_i - tw_i^s}{tw_i^e - tw_i^s}$$

f_1: the failure rate of agile satellite observation requests;

f_2: the timeliness of scheduled requests, where $\frac{t_i - tw_i^s}{tw_i^e - tw_i^s}$ represents the timeliness of request r_i, and normalization is applied here by dividing N.

The proposed bi-objective function can not only meet the demand to obtain reasonable schedules with considerable profits, but also potentially help observations to be accomplished as soon as possible within their VTW, which is quite meaningful for practical application and can provide the coming observation tasks with more opportunities to be fulfilled.

2.4. Constraints

Here are the constraints which define the feasible sequence of MO-AEOSSP:

$$t_i \geq tw_i^s,\ \ if x_i = 1 \tag{1}$$

$$t_i + du_i \leq tw_i^e,\ \ if x_i = 1 \tag{2}$$

$$t_i + du_i + tr_{ij} \leq t_j,\ \ if x_i, x_j = 1 \tag{3}$$

$$\sum_{i=1}^{N} x_i du_i \leq \alpha M \tag{4}$$

Constraints (1) and (2) restrict that an observation window must be included in its corresponding VTW, where t_i and $(t_i + du_i)$ are the start and end time for observing r_i;

Constraint (3) indicates that the observation windows of any two observations cannot overlap and there must be enough transition time for the satellite to maneuver the observation angle from the former request r_i to the adjacent request r_j, since a satellite can only support one observation at any given moment.

Constraints (4) states that the total occupied storage for scheduled requests cannot exceed the total onboard memory, where M denotes the onboard memory and α is a conversion parameter of memory from duration.

3. Deep reinforcement learning and parameter transfer based approach (RLPT) for MO-AEOSSP

To resolve the multi-objective optimization of the request failure and the average timeliness in MO-AEOSSP, RLPT and some specific strategies are presented in this section. The framework of RLPT is first introduced, then the MO-AEOSSP is regarded as the test problem, the architecture of the corresponding neural network and the training method are illustrated in the remaining part of this section.

3.1. Framework

Fig. 2 shows the framework of the proposed RLPT. Similarly as [26], the test problem is first converted into a series of scalar optimization sub-problems through a weighted sum approach [49]. Unlike the previous work, the first and the last sub-problems are selected as initial sub-problems after decomposition, then their corresponding neural networks are trained through DRL methods. After initialization, a neighborhood-based parameter transfer strategy is adopted at every training step before termination, where the parameters of trained networks are transferred to the networks of neighborhood sub-problems in order to accelerate the training speed. After all the corresponding networks have been trained, only a simple feed-forward process is needed for producing the approximate PF of MO-AEOSSP.

Let $\boldsymbol{\lambda}_i = (\lambda_{i1}, \ldots, \lambda_{im})^m$ be a weight vector which forms the convex combination of m different objectives, then the MO-AEOSSP can be decomposed into l sub-problems by a set of uniformly distributed weight vectors $\Lambda = (\boldsymbol{\lambda_i}, \ldots \boldsymbol{\lambda_l})$. Each sub-problem is solved using neural network and the approximate PF can be generated by solving each scalar optimization problem [50] as follows:

$$\min g^{ws}(x|\boldsymbol{\lambda_i}) = \sum_{j=1}^{m} \lambda_{ij} f_j(x) \tag{5}$$

where $g^{ws}(x|\boldsymbol{\lambda_i})$ is the objective function of the ith sub-problem using $\boldsymbol{\lambda_i}$ as a coefficient vector, f_j is the jth objective, m denotes the number of objectives and x are the variables to be optimized.

It is obvious from Eq. (5) that adjacent sub-problems share similar features and have close optimal solutions. In order to alleviate the influence of unidirectional parameter transfer, the corresponding networks of both the first and the last scalar problems are trained at the very beginning and in the following training procedure the best parameters θ_i, θ_{l-i+1} of the i^{th} and $(l-i+1)^{th}$ networks are set as initial trainable parameters for the network training in the $(i+1)^{th}$ and $(l-i)^{th}$ networks. The aforementioned strategy is depicted in Fig. 3.

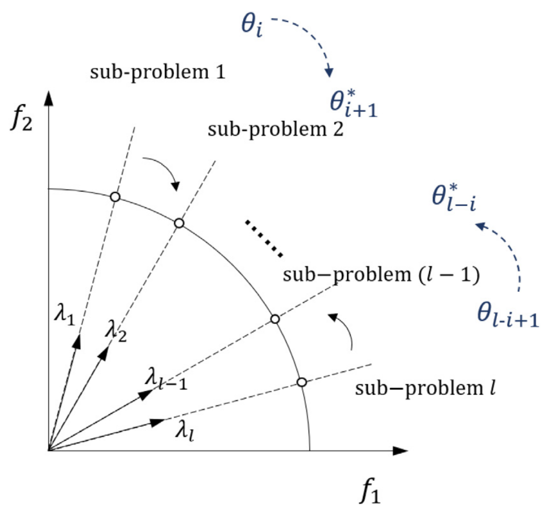


Fig. 3. The parameter transfer strategy for RLPT. Assume the i^{th} and $(l-i+1)^{th}$ networks have been trained and their optimized parameters are denoted by θ_i and θ_{l-i+1}. Then θ_i and θ_{l-i+1} are set as the starting points of the networks training in the $(i+1)^{th}$ and $(l-i)^{th}$ sub-problems (represented by θ^*_{i+1} and θ^*_{l-i}), respectively. The angle between the sub-problems and the radius have no actual meaning, they are just used with illustrative purposes.

3.2. Architecture of the proposed neural network

In this section, the main contributions of this paper are proposed: the neural network for the sub-problem of MO-AEOSSP and the scheduling strategy for sub-problem.

Given a set of input requests $R = \{r^i, i = 0, 1, \ldots, M\}$, where M is the number of requests and r^0 is a virtual request set to denote the start-moment of the scheduling horizon. Each r^i consists of a sequence of tuples $\{r^i_t = (s^i, d^i_t), t = 0, 1, \ldots\}$, where s^i and d^i_t are the static and dynamic elements of the input request r^i at decoding time t. Starting from selecting $r^0 \in R_0$ as the first input y_0, the decoded sequence can be denoted by $Y_t = \{y_0, y_1, \ldots, y_t\}$ and R_t is the available candidate inputs at every decoding step $t(t = 1, 2, \ldots)$, then $y_{t+1} \in R_t$ is chosen as the input request of the next decoding moment before R_{t+1} and the set of dynamic elements $D_{t+1} = \{d^i_t, i = 0, 1, \ldots, M\}$ are updated according to the problem constraints. The above procedure continues until the termination criteria is satisfied and the probability chain rule is adopted here to decompose the probability of generating $Y_t = \{y_0, y_1, \ldots, y_{t'}\}$:

$$P(Y|R_0;\theta) = \prod_{t=0}^{t'} P(y_{t+1}|Y_t, R_t;\theta) \tag{6}$$

$$R_{t+1} = f(y_{t+1}, R_t) \tag{7}$$

where θ denote the learnable parameters, $f(\cdot)$ is the function used to update the available candidate inputs R_{t+1} after y_{t+1} is chosen.

With Eq. (6) and (7), the probability distribution of potentially selected requests in the next decoding step can be computed. An attention-based neural network is used as the policy which generates the sequence Y in a way that minimizes the sub-problem objective.

The parameters θ are obtained by maximizing the conditional probability of the training set:

$$\theta = argmax \sum_{Y,R_0} logp(Y|R_0;\theta) \tag{8}$$

The architecture of the network is shown in Fig. 4, which is basically a sequence-to-sequence model with an encoder–decoder structure. In this work, each network consists of three parts, feature encoder, task encoder and decoder. Details are elaborated below.

3.2.1. Feature encoder

As illustrated in Fig. 4, the feature encoder is used to extract features of input requests R_t by an embedding layer at the recursive step t, encoding inputs to a high-dimensional vector space. In the sub-problem of MO-AEOSSP, the feature encoder determines both the static and the dynamic information, including the following features:

(1) Static features: $s^i = \{tw^s_i, tw^e_i, du_i, p_i\}$, including the starting and ending time of VTW, duration and priority of r_i according to Section 2;

(2) Dynamic features: $d^i_t = \{state^i_t, tc_t, rs_t\}$, where the $state^i_t$ indicates whether r_i can be selected, tc_t is the starting scheduling moment of the decoding step t, rs_t represents the remaining onboard storage, respectively.

Since the input order shows no sequential information and is not meaningful, positional encoding is unnecessary for the feature encoder, thus a 1-dimensional convolution (Conv-1D) layer is used rather than a recurrent neural network (RNN) encoder to alleviate the computational complications without decrease the efficiency. The number of in-channels equals the dimension of inputs. For instance, the numbers of channels and filters are set to 4 and D for static features.

3.2.2. Task encoder

The task encoder is employed to provide the decoder with the information of the decoded sequence and decision-making process. Owing to the requirement of memorizing previous steps, an RNN is needed for modeling the task encoder. Gated Recurrent Unit (GRU) [51], a simplified variant of Long Short-Term Memory (LSTM) [52], is adopted in this work. Let h_t be the hidden state of the GRU cell which stores the knowledge of previous steps at decoding time t and e_{s_j} be the embedded input of last selected request which is generated by the feature encoder, then h_t and e_{s_j} are calculated together to output o_t for the decoder to make decision for the next step. Noticeably, only the embedded static features e_{s_j} are used in the task encoder, the dynamic elements $e_{d^j_t}$ are used and updated in the decoder.

3.2.3. Decoder

The decoder makes decision based on the outputs of the feature encoder and task encoder, pointing to an embedded input at every decoding step. The chosen request is then taken as the input of the task encoder for the next decoding. A modified attention mechanism is used during the decision-making process and two attention layers are designed in this section. As Fig. 4 shows, the first attention layer calculates the attention distribution while the second layer combines the given attention information, static and dynamic embeddings to produce the probability distribution of unselected requests.

The output attention distribution from the first layer is denoted by an alignment vector a_t. Given the embedded static, dynamic input e_{s_j}, $e_{d^j_t}$ and the output conditional vector of task encoder o_t, a_t is then computed by the following function:

$$a_t = a_t(e_s, e_{d_t}, o_t) = \text{softmax}(v_a^T \tanh(W_a[e_s; e_{d_t}; o_t])) \tag{9}$$

where "[A;B]" represents the concatenation of two vectors, v_a and W_a are trainable parameters.

The output of the former layer, is put into the second layer together with the static embeddings to calculate a context vector c_t:

$$c_t = \sum_{i=1}^{n} a^i_t e_{s_i} \tag{10}$$

where n is the number of unselected requests at t.

The conditional probabilities are computed and normalized as follows:

$$P(y_{t+1}|Y_t, R_t) = \text{softmax}(v_c^T \tanh(W_c[e_t; c_t])) \tag{11}$$

where $e_t = [e_s; e_{d_t}]$, v_c and W_c are trainable parameters.

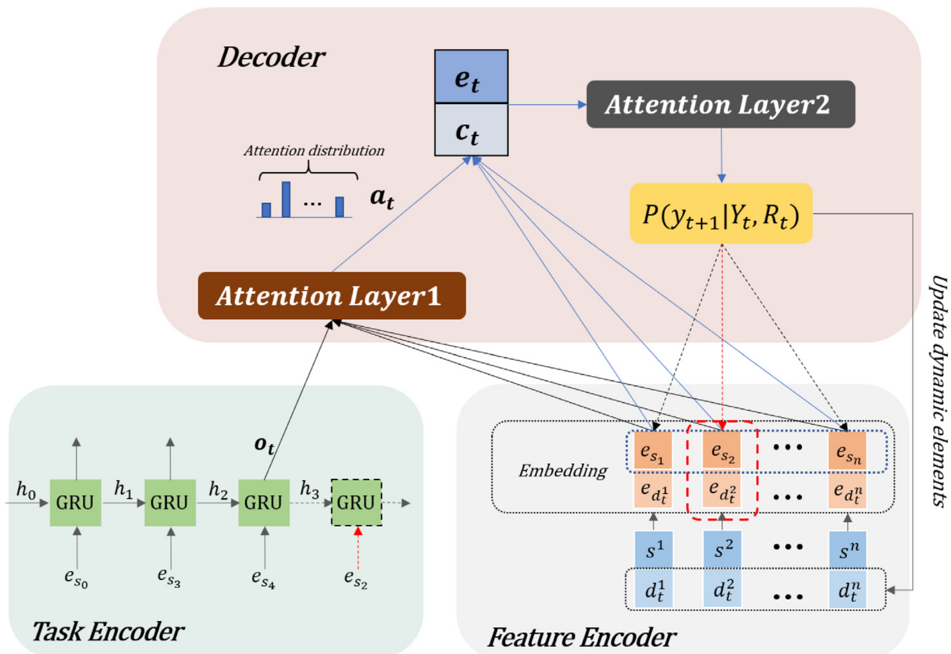


Fig. 4. The three modules of the sub-problem neural network. e_{s_i} and $e_{d_t^i}$ are the embeddings mapped by the input static and dynamic features s^i, d_t^i. h_t and o_t are the hidden state and the output conditional vector of GRU respectively. The output vector of the first attention layer is denoted by a_t. The context vector c_t and e_t (the concatenation of e_s and e_{d_t}) are the inputs of the second attention layer. This figure is a detailed view of the neural network modeling block shown in Fig. 2.

3.3. Training method

An Actor–Critic training algorithm [53] is adopted for training the proposed neural network, as presented in Algorithm 1. The training algorithm contains two networks:

(1) Actor network: the neural network for scheduling the sub-problem, producing a probability distribution for deciding the next action at every decoding step. The structure of the actor network is fully illustrated in Section 3.2.

(2) Critic network: the network which evaluates reward for a specific state of any sub-problem instance. The critic network employs four Conv-1D layers, parameters of each layer are presented in Table 2 of Section 4.2.

Algorithm 1 Actor-critic training

Input: parameters of Actor and Critic network θ, θ_c
    N problem instances generated from distributions ϕ
Output: the optimal parameters θ, θ_c

```
initialize actor network params θ
initialize critic network params θ_c
for iteration ← 1, 2, ··· do
  reset gradients : dθ ← 0, dθ_c ← 0
  for k = 1, 2, ..., N do
    step counter t ← 0
    while not terminated do
      choose next request y^k_(t+1) according to the distribution
        P(y^k_{t+1} | Y^k_t, R^k_t)
      Update new state R^k_{t+1} to R^k_t
      t ← t + 1
    end while
    compute reward R̃^k
  end for
  dθ ← (1/N) Σ_{k=1}^{N} (R̃^k − V(R^k_0; θ_c)) ∇_θ log P(Y^k | R^k_0)
  dθ ← (1/N) Σ_{k=1}^{N} ∇_{θ_c} (R̃^k − V(R^k_0; θ_c))^2
  θ = Adam(θ, dθ)
  θ_c = Adam(θ_c, dθ_c)
end for
```

$$d\theta \leftarrow \frac{1}{N}\sum_{k=1}^{N}(\widetilde{R}^k - V(R_0^k;\theta_c))\nabla_\theta \log P(Y^k|R_0^k)$$

$$d\theta \leftarrow \frac{1}{N}\sum_{k=1}^{N}\nabla_{\theta_c}(\widetilde{R}^k - V(R_0^k;\theta_c))^2$$

4. Computational experiment

In this section, the method for generating training data, parameter settings and performance metrics are first introduced. Then computational experiments are carried out and the performance of the proposed approach is evaluated.

The comparison algorithms are programmed in the Evolutionary Multi-objective Optimization Platform (PlatEMO) [54], the experiments are conducted using a single GPU RTX 2080-Ti, and a i7-9700K CPU with 64 GB RAM. The DRL framework embedded in Pytorch 1.2 coded in Python 3.7 is used in this study.

4.1. Dataset

Due to the lack of common benchmark and the requirement of large amounts of data for training the RLPT network, we constructed training instances by a similar method used by Liu et al. [13].

Liu et al. generated their test instances using a uniform random distribution through different geographical regions and each request is defined by latitude, longitude and altitude. In order to train the neural network, modification of the generating method is adopted in this paper. The scheduling horizon is set to [0, 100] in seconds and the geographical position of each request is reflected as a VTW in the observation horizon. The VTW of each task is generated among [3.4, 5.6] in seconds according to the practical rules of satellite observation. The duration and transition time between two adjacent requests are set to [0.5, 1.1] and 0.8 in seconds while the profit is in the range of [1, 10]. The total memory is set to 1. All the aforementioned elements follow

Table 2
Main parameters of the Actor–Critic network.

Actor Network
Feature Encoder: Conv-1D(D_{input_size}, Filter = 128, kernel size = 1, stride = 1) Task Encoder: GRU(hidden size = 128, layers = 1, dropout = 0.1) Decoder: Attention(No hyper parameters)
Critic Network
Encoder: Conv-1D(D_{input_size}, Filter = 256, kernel size = 1, stride = 1) Conv-1D(256, Filter = 20, kernel size = 1, stride = 1) Conv-1D(20, Filter = 20, kernel size = 1, stride = 1) Conv-1D(20, Filter = 1, kernel size = 1, stride = 1)

the specific distribution ϕ and the uniform random distribution is adopted to generate datasets in this paper. Besides, all the training data have to be normalized before being used for training.

We train the RLPT networks using 100,000 instances of 60- and 80-request MO-AEOSSPs respectively and MO-AEOSSPs of 50–200 tasks are tested on the trained networks. The initial networks are trained for 5 epochs while others use the instances in training for 2 epochs because of the applied parameter transfer method.

The dataset towards MO-AEOSSP, which contains various-size instances for training and test, together with the source code are publicly available at https://github.com/wlnelysion/MO-AEOSSP. Instances of 50, 60, 80, 100, 120, 150, 200 requests are organized into groups in the repository, 30 instances for each scale. Different instances are denoted by "Size_NO.", where "Size" is the number of input requests in this instance and "NO." is the index. For example, "100_1" means the first instance of the 100-request MO-AEOSSP.

Because of the length limit, we cannot present the evaluation of all the instances in this study. Instead, one of the representative instances for each size is selected and all the chosen instances have been collected in a specific folder (named EvaRLPT) in the repository.

4.2. Parameter settings

The parameter settings of the Actor–Critic network are shown in Table 2. We use the Adam Optimizer [55] with a peak learning rate of $5e-5$ to train both the actor and critic networks. Xavier initialization strategy [56] is employed to initialize the networks of the first two sub-problems and parameters of the following networks are generated by parameter transfer strategy.

The performance of the proposed network is compared with the results of three major types of MOEAs:

(1) NSGA-II [57], a well-known domination-based MOEA;
(2) MOEA/D [50], a well-known decomposition-based MOEA;
(3) IBEA [58], a well-known metric-based MOEA.

Parameter settings for the algorithms are set as below:

- All the comparative algorithms apply the coding and decoding strategies presented by Yuan et al. [59];
- For the variation procedure, the order crossover operator and the insertion mutation operator are adopted, which are applied in [59] before;
- For the individual selection, the multi-objective binary tournament selection operator is used to form the mating pool;
- The comparative MOEA/D use the PBI approach as its decomposition strategy with a T (the size of neighborhood) of 10, which is also employed in [50];
- The IBEA applies the fitness assignment operator as [58] does and the scaling factor is set to 0.05;
- The population size is 100 for all the comparative algorithms;
- The maximum number of evaluations for all the MOEAs 5000, 10 000 and 20 000 respectively. By preliminary experiments, we figure out that MOEAs with more than 20 000 evaluations do not show obviously better convergence than those with 20 000 evaluations but need much more computing time. Thus, in this study only MOEAs with no more than 20 000 evaluations are considered;
- Different algorithms are denoted by "A-NO.", where "A" is the specific algorithm and "NO." is the number of evaluations, for instance "NSGAII-5000" means the NSGA-II with 5000 evaluations;
- The number of sub-problems is set to 100.

4.3. Performance metrics

The following methods and metrics are used to evaluate the performance of algorithms in this work:

(1) The pareto front (PF). The PF provide an intuitive view of the non-dominated solutions reserved by each algorithm.

(2) The hypervolume (HV) metric [60], which is the volume in the target space enclosed by the non-dominated solutions and the reference point. Since a reference point is needed when calculating the HV metric, we use the worst possible values (1,1) of the two objectives as the reference point.

(3) The mean computing time. The difference of computational cost between approaches is compared based on the combination of the mean computing time and other metrics including the PF and the HV.

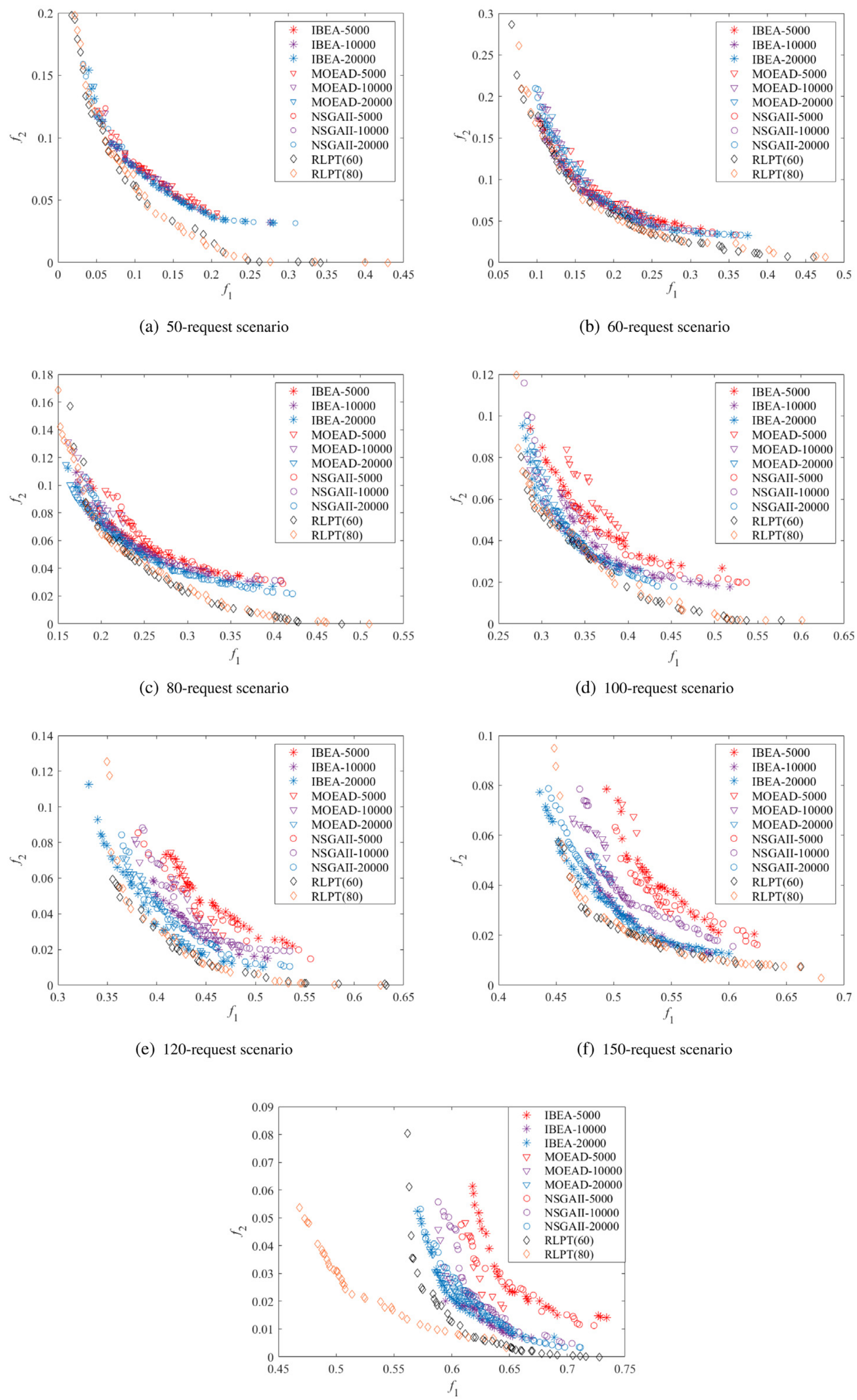
(4) The number of non-dominated solutions for RLPT. To further clarify the impact of using various-size training data on network training, the numbers of solutions reserved in the final PFs of RLPTs are analyzed.

4.4. Experimental results

In this subsection, instances with 50–200 requests are generated using the distribution ϕ which is also applied by the generator of training data. For each size, RLPT networks trained on 60 and 80 requests, which are donated by RLPT(60) and RLPT(80), and the above-mentioned comparative MOEAs are run 30 times to address the MO-AEOSSPs. The PFs which reserve only the non-dominated solutions after a random run of different instances using different algorithms are depicted in Fig. 5, where the objective value of f_1(the failure rate) is represented by x-axis and y-axis denotes f_2(the timeliness of scheduled requests).

It is obvious that in all the instances, RLPTs shows better domination than the comparative MOEAs. Regardless of instance size, RLPTs outperform all the comparison algorithms in obtaining non-dominated solutions and preserving the quality of spreading and diversity.

For scenarios of small instance size, i.e. 50-, 60-, 80-request MO-AEOSSPs, all the traditional methods can address the corresponding problems within a small number of iterations, which can be easily found out through Figs. 5(a)–5(c) since the non-dominated solutions obtained by comparison methods of different evaluations can almost overlap with each other. However, maintaining the same request failure of MO-AEOSSP, the proposed RLPTs can produce more qualified solutions with better performance of timeliness in comparison with the other comparative methods, and vice versa. Furthermore, regarding the distribution of solutions, the solutions of both RLPT(60) and RLPT(80) are well-distributed in x-axis and y-axis, thus, the two different RLPTs are of about the same capability of spreading. It can be regarded that RLPT occupy the first place of the ability of distributing, followed by NSGA-II and IBEA while MOEA/D performs worst.



(a) 50-request scenario

(b) 60-request scenario

(c) 80-request scenario

(d) 100-request scenario

(e) 120-request scenario

(f) 150-request scenario

(g) 200-request scenario

Fig. 5. Non-dominated solutions reserved in different scenarios of MO-AEOSSP using the proposed RLPTs (trained using instances of 60 and 80 requests) and different comparison algorithms with 5000, 10 000 and 20 000 evaluations. (f_1 denotes the failure rate and f_2 denotes the timeliness of scheduled requests.).

When comes to scenarios of 100-, 120-, 150- and 200-request, MOEAs with different iterations possess quite different ability of convergence because of the larger solution space. Obviously, a larger number of iterations can lead to more superior performance of MOEAs, but it also needs more computing time, which will be discussed later. Still in these scenarios, RLPT(60) and RLPT(80) are able to outperform the other methods in accordance to the PFs in Figs. 5(d)–5(g), exhibiting significantly enhanced capability of spreading and convergence even when compared with NSGA-II-20000, MOEA/D-20000 and IBEA-20000. The PFs of all the addressing methods are not such well-distributed in these instances, since more request conflicts occur because of the overlap of VTWs caused by the increasing number of requests, which further leads to the difficulty for guaranteeing the timeliness of scheduled requests. It is worth noting that among these instances RLPT(80) shows a better ability for getting high-quality solutions and preserving extreme solutions than RLPT(60), especially when solving the scenario of 200-request MO-AEOSSP. At the same extent of timeliness, RLPT(80) is capable of producing solutions with a lower rate of request failure than other methods.

Fig. 6 shows the mean computing time of different algorithms. We can easily figure out that all the comparison methods with 5000 evaluations seem very efficient for addressing MO-AEOSSPs, occupying almost all the best position of computing time for all the instance size except 200. However, when combined with the PFs in Fig. 5, their efficiency turns to be rather questionable for that although they are able to conduct solutions within a very short period, the quality of some PFs is not acceptable, especially for the large-scale situation. Apart from the 5000-evaluation methods, the proposed RLPTs are very competitive when compared to other comparative MOEAs with 10 000 and 20 000 evaluations. According to Figs. 5 and 6, increasing the number of evaluations can definitely enhance the performance of NSGA-II, MOEA/D and IBEA, but more computing time is needed for iteration. We find out that the computing cost of RLPT(60) and RLPT(80) does not have an apparent tendency related to the scale of the instance while other comparison algorithms show a clear linear correlation with the number of evaluations and instance size.

Table 3 states the average HV values of all the methods. As RLPTs always get the highest HVs which imply the best overall performance, the results illustrate that RLPTs generally outperform all the comparison MOEAs and present considerable generalization ability. More precisely, RLPT(60) performs best on the MO-AEOSSPs of 50 and 60 requests while RLPT(80) exhibits the best on the remaining instances. For the comparison MOEAs, there is no doubt that MOEAs can get their highest HVs after 20 000 evaluations, and some MOEAs can be quite competitive under certain instance scales such as 120- and 150-request MO-AEOSSPs. But when combined with the previous analysis of the computation time and the obtained PFs, the comparison MOEAs present more inferior comprehensive performance than RLPTs. In terms of HVs of RLPTs, basically the performance difference between RLPT(60) and RLPT(80) is not such significant except for the case of 200 requests, even though they seem good at dealing with different instances. In addition, the overall trend for the HV of all the algorithms states that the HV decrease as the size of the candidate requests increases. This is mainly because that the raising of the instance scale makes the candidate requests set of MO-AEOSSP larger, which complicates the problem, leading to a larger solution space and more severe request conflicts. But the scheduling horizon for request arranging is fixed, which makes the percentage of tasks that can be scheduled gradually decrease, further causes the decline of HV and presents an inverse relationship between the HV and the instance size.

Finally, the impact of RLPTs trained by instances of different sizes is analyzed by comparing the overall performance of RLPT(60) and RLPT(80), calculation cost and metrics of the obtained solutions. Table 4 indicates the number of solutions reserved in the final PFs of RLPTs. We are not surprised to find that not all the solutions generated by RLPT(60) and RLPT(80) are non-dominated. All of the above information states that these two methods perform basically the same on all scales, proving that the RLPT proposed in this article is of great ability of generalization and scalability so it can be used as an efficient tool to address the MO-AEOSSP.

Table 3
The average HVs of algorithms in scenarios of different instance size (30 runs for each algorithm). The bolded value in each column is the highest average HV value of each scenario.

Method	Instance size						
	50	60	80	100	120	150	200
RLPT(60)	**0.9723**	**0.9159**	0.8429	0.7430	0.6737	0.5833	0.4879
RLPT(80)	0.9694	0.9088	**0.8538**	**0.7472**	**0.6786**	**0.5879**	**0.5739**
NSGA-II-5000	0.9162	0.8486	0.7956	0.6852	0.6249	0.5404	0.4372
NSGA-II-10000	0.9208	0.8637	0.8132	0.7033	0.6460	0.5656	0.4553
NSGA-II-20000	0.9234	0.8749	0.8246	0.7284	0.6650	0.5701	0.4746
MOEA/D-5000	0.9065	0.8379	0.7956	0.6809	0.6170	0.5410	0.4402
MOEA/D-10000	0.9124	0.8550	0.8066	0.7033	0.6439	0.5643	0.4633
MOEA/D-20000	0.9224	0.8604	0.8144	0.7192	0.6621	0.5801	0.4724
IBEA-5000	0.9147	0.8542	0.8012	0.7057	0.6272	0.5401	0.4355
IBEA-1000	0.9242	0.8697	0.8119	0.7253	0.6403	0.5682	0.4498
IBEA-20000	0.9279	0.8701	0.8293	0.7362	0.6700	0.5788	0.4717

Table 4
The numbers of non-dominated solutions obtained by different RLPTs. To compare performance of RLPTs trained by instances of different scales and to further illustrate the generalization ability of the proposed approach in this paper, only the results of RLPT(60) and RLPT(80) are presented in this table.

Method	Instance size						
	50	60	80	100	120	150	200
RLPT(60)	34	49	45	34	30	35	41
RLPT(80)	40	40	52	41	34	45	38

5. Conclusion and future work

In this paper, we consider a multi-objective agile earth observation satellite scheduling problem (MO-AEOSSP) where two objectives, namely the failure rate of observation requests and the timeliness of scheduled requests, are optimized simultaneously, which is a highly complicated NP-hard problem. Traditional iteration-based methods such as MOEAs designed for this problem are usually quite time-consuming. These methods also lack adaptability such that they may perform unexpectedly badly for unseen problem instances. To address these issues, and motivated by recent research focusing on applying deep reinforcement learning (DRL) to solving combinatorial optimization problems, a deep reinforcement learning and parameter transfer based approach (RLPT) is proposed in this study. Encoder–decoder structure neural networks are adopted by RLPT to address single objective sub-problems which are decomposed from MO-AEOSSP through a weight sum approach. RLPT applies the DRL method to training all the networks and a parameter transfer strategy is incorporated into RLPT to accelerate the training speed. Three well-known MOEAs: NSGA-II, MOEA/D and IBEA are implemented and examined to evaluate the merits of the proposed RLPT.

Experimental results illustrate that RLPTs are able to get high-quality solutions with competitive performance in terms of convergence, spreading and computing efficiency in comparison with standard MOEAs. Let the RLPT trained on a set of n-request instances be RLPT(n). Our conclusions can be summarized as follows. (1) the pareto fronts (PFs) generated by RLPT(60) and RLPT(80) show great domination among instances of other sizes; (2) by reference to the quality of approximate PFs and their spreading, RLPTs outperform MOEAs; (3) when referring to the hypervolume (HV) indicator and computing time, we conclude that RLPTs are capable to solve MO-AEOSSP within a quite acceptable short time without sacrificing the quality of solutions; (4) RLPT(60) and RLPT(80) show almost the same level of solution quality, HV values and computing time on all tested instances, revealing their excellent generalization and scalability. Once a well-trained RLPT is acquired, it can be applied to various-size

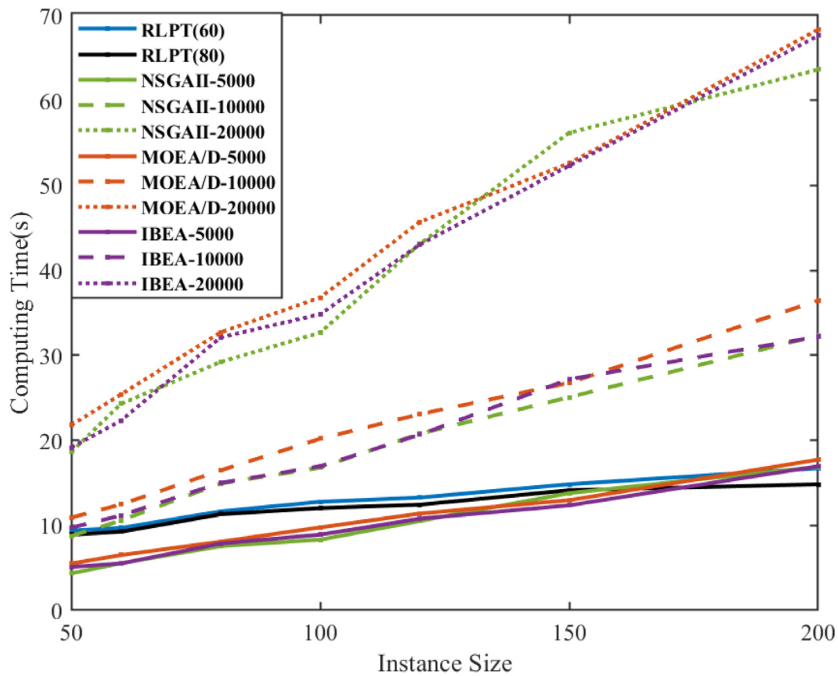


Fig. 6. The computing time (s) on instances of variable size using different algorithms.

instances and the quality of the obtained approximate PFs is satisfying. In this regard, the proposed RLPT is expected to produce good results for MO-AEOSSP in practical situations.

For future work, we would like to further optimize the architecture of the RLPT network, by considering more important features of the MO-AEOSSP, seeking to further improve the performance of RLPT and make it more available for practical application.

CRediT authorship contribution statement

Luona Wei: Conceptualization, Methodology, Software, Writing - original draft. **Yuning Chen:** Formal analysis, Investigation. **Ming Chen:** Writing - review & editing, Validation. **Yingwu Chen:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work is partially supported by the National Natural Science Foundation of China, Grant 71701203, 72001212 and Natural Science Foundation of Hunan Province, China, Grant No. 2018JJ3618.

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